

STAT 525

Chapter 7

General Linear Test and Multicollinearity

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General Linear Test

- Comparison of a **full** model and **reduced** model that involves a subset of full model predictors (i.e., hierarchical structure)
- Involves a comparison of unexplained SS
- Consider a full model with k predictors and reduced model with l predictors ($l < k$)

- Can show that under null hypothesis
$$F^* = \frac{(\text{SSE}(R) - \text{SSE}(F))/(k - l)}{\text{SSE}(F)/(n - k - 1)} \sim F \text{ distribution}$$

df₁
→ predictor difference
df₂
p: parameters

- Degrees of freedom for F^* are the number of **extra** variables and the error degrees of freedom for the full model

- Testing the Null hypothesis that the regression coefficients for the **extra** variables are all zero.
- Examples:
 - X_1, X_2, X_3, X_4 vs $X_1, X_2 \rightarrow H_0 : \beta_3 = \beta_4 = 0$ if we are sure that $\beta_5, \dots$ are exactly 0's *Assuming all other β 's = 0*
 - X_1, X_2, X_4 vs $X_1 \rightarrow H_0 : \beta_2 = \beta_4 = 0$ if we are sure that $\beta_3, \beta_5, \dots$ are exactly 0's
 - X_1, X_2, X_3, X_4 vs $X_1 \rightarrow H_0 : \beta_2 = \beta_3 = \beta_4 = 0$ if we are sure that $\beta_5, \dots$ are exactly 0's
- Because $SSM + SSE = SSTO$, can also compare using explained SS (SSM)

Extra SS and Notation

- Consider $H_0 : X_1, X_3$ vs $H_a : X_1, X_2, X_3, X_4$
 H_0 : only 2 predictors needed
- Null can also be written $H_0 : \beta_2 = \beta_4 = 0$
- Write $SSE(F)$ as $SSE(X_1, X_2, X_3, X_4)$
- Write $SSE(R)$ as $SSE(X_1, X_3)$
- Difference in SSE's is the extra SS
- Write as

$$SSE(X_2, X_4 | X_1, X_3) = SSE(X_1, X_3) - SSE(X_1, X_2, X_3, X_4) \checkmark$$

- Recall SSM can also be used

$$\begin{aligned} SSM(X_2, X_4 | X_1, X_3) &= SSM(X_1, X_2, X_3, X_4) - SSM(X_1, X_3) \implies \\ SSM(X_1, X_2, X_3, X_4) &= SSM(X_1, X_3) + SSM(X_2, X_4 | X_1, X_3) \end{aligned}$$

General Linear Test in Terms of Extra SS

- Can rewrite F test as

$$\checkmark \quad F^* = \frac{\text{SSE}(X_2, X_4 | X_1, X_3) / (4 - 2)}{\text{SSE}(X_1, X_2, X_3, X_4) / (n - 5)}$$

- Under H_0 , $\checkmark \quad F^* \sim F(2, n - 5)$; under H_1 , $F^* \sim$ noncentral $F(2, n - 5)$

- If reject, conclude either X_2 or X_4 or both contain additional useful information to predict Y in a linear model with X_1 and X_3
 Reject $H_0: \beta_2$ and $\beta_4 \neq 0$

- Example: Consider predicting GPA with HS grades, do SAT scores add any useful information?
- If neither H_0 nor H_1 is correct, p-value has no rigorous statistical meaning, but still serves as a comparison tool.

Special Cases

- Consider testing individual predictor X_i based on

$$\text{SSE}(X_i | X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{p-1})$$

- These are related to SAS's indiv parameter t -tests

$$\downarrow F(1, n - p) = t^2(n - p)$$

- Can decompose SSM variety of ways

- $\downarrow$ – Decomposition of $\text{SSM}(X_1, X_2, X_3)$ *Depend on order they enter the model*

$$= \text{SSM}(X_1) + \text{SSM}(X_2 | X_1) + \text{SSM}(X_3 | X_2, X_1)$$

$$= \text{SSM}(X_2) + \text{SSM}(X_1 | X_2) + \text{SSM}(X_3 | X_2, X_1)$$

$$= \text{SSM}(X_3) + \text{SSM}(X_2 | X_3) + \text{SSM}(X_1 | X_2, X_3)$$

- Stepwise sum of squares called Type I SS

Type I SS and Type II SS

- Type I and Type II are very different

order – Type I is sequential, so it depends on model statement

no order – Type II is conditional on all others, so it does not depend on model statement

- For example, model $y = x_1 \ x_2 \ x_3$ yields

Step-wise: In order of appearance

Type I	Type II
SSM(X_1)	SSM($X_1 X_2, X_3$)
SSM($X_2 X_1$)	SSM($X_2 X_1, X_3$)
SSM($X_3 X_1, X_2$)	SSM($X_3 X_1, X_2$)

All 3 SSMs account for all predictors

- Could variables be explaining same SS and “canceling” each other out, such that we need to cautions about testing results? *Yes. If variables are correlated. (Multicollinearity)*

- *Type II may be inflated*

Always check PROC CORR before testing

Example: Body Fat (p.256)

- Twenty healthy female subjects
- Y is body fat via underwater weighing
- Underwater weighing is expensive/difficult
- X_1 is triceps skinfold thickness
- X_2 is thigh circumference
- X_3 is midarm circumference

All 3 correlated... people with ↑ fat
tend to be fat in all areas.

- Investigate the model with all three predictors:

```
data a1;
  infile 'U:\Ch07ta01.txt';
  input skinfold thigh midarm fat;

proc reg data=a1;
  model fat=skinfold thigh midarm /ss1 ss2;
run;
```

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	396.98461	132.32820	21.52	<.0001
Error	16	98.40489	6.15031		
Corrected Total	19	495.38950			

All 3 together is significant

Root MSE	2.47998	R-Square	0.8014
Dependent Mean	20.19500	Adj R-Sq	0.7641
Coeff Var	12.28017		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	117.08469	99.78240	1.17	0.2578
skinfold	1	4.33409	3.01551	1.44	0.1699
thigh	1	-2.85685	2.58202	-1.11	0.2849
midarm	1	-2.18606	1.59550	-1.37	0.1896

Individual not significant

Conclusions

- Set of three variables helpful in predicting body fat ($P < 0.0001$)
- None of the individual parameters is significant
 - Addition of each predictor to a model containing the other two is not helpful
 - Example of multicollinearity
 - Will discuss more in next topic
- Will now focus on extra SS

High Drop Type 1 SS to Type 2:
Multicollinearity

- Output Using SS1 & SS2 gives an additional two columns

Parameter Estimates

Variable	DF	Parameter Estimate	(Sequential) Type I SS	(Adjusted / Partial) Type II SS
Intercept	1	117.08469	8156.76050	8.46816
skinfold	1	4.33409	352.26980	12.70489
thigh	1	-2.85685	33.16891	7.52928
midarm	1	-2.18606	11.54590	11.54590

Once adjusting for thigh and midarm in the model, skinfold no longer explains that much "new" information.

More than 90% of Type I SS of skinfold can also be explained by thigh and midarm

skinfold gets credit for everything it explains first: even parts that overlap with thigh and arm.

• High value from what thigh and midarm explained.

skinfold alone is already highly significant

- Investigate the model via general linear tests: fat=skinfold

```
proc reg data=a1;  
    model fat=skinfold;  
run;
```

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	352.26980	352.26980	44.30	<.0001
Error	18	143.11970	7.95109		
Corrected Total	19	495.38950			
Root MSE		2.81977	R-Square	0.7111	
Dependent Mean		20.19500	Adj R-Sq	0.6950	
Coeff Var		13.96271			

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-1.49610	3.31923	-0.45	0.6576
skinfold	1	0.85719	0.12878	6.66	<.0001

- Skinfold now helpful. Note the change in coefficient estimate and standard error compared to the full model.

- Does this variable alone do the job?
- Perform general linear test

```
proc reg data=a1;
  model fat=skinfold thigh midarm;
  thimid: test thigh, midarm;
run; quit;
```

Test if $H_0: \beta_{thigh} = \beta_{midarm} = 0$, to see if skinfold is the only predictor needed.

Test thimid Results for Dependent Variable fat

Source	DF	Mean Square	F Value	Pr > F
Numerator	2	22.35741	3.64	0.0500
Denominator	16	6.15031		

- Appears there is additional information in the variables. Perhaps the addition of one more variable would be helpful.
- Adding thigh and midarm may slightly improve the model.*

Unique strength of relationship of X v.s. Y
after removing other X 's.
✱ Unique Contributions ✱

Partial Correlations

↑↑↑ : X_i still explains in Y after other
adjusting for other variables.
↓↓↓ : X_i adds little new information.

Quantifies unique explanatory contributions after
controlling for other variables.

- Measures the strength of a linear relation between two variables taking into account other variables or after adjusting for other variables, while marginal t-test measure the existence of such a linear relationship
- Procedure for X_i vs Y Ex: thigh v.s. fat
 - Predict Y using other X 's Ex: Predict Y (fat) using all other X 's (skinfold and midarm) → get residuals for fat
 - Predict X_i using other X 's Ex: Predict X_i (thigh) using all other X 's (skinfold and midarm) → get residuals for thigh
- Each residual represents what is not explained by the other variables Additional info.
 - The correlation between residuals is the partial correlation
 - The regression coefficient residuals is the regression coefficient of X_i in the full model
 - The test of association is the marginal t test in the full model
- Looking for additional information in X_i that better explains Y

Example: Body Fat

```
proc reg data=a1;
    model fat=skinfold thigh midarm / pcorr2;
run;
```

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Squared Partial Corr	Type II
Intercept	1	117.08469	99.78240	1.17	0.2578	.	.
skinfold	1	4.33409	3.01551	1.44	0.1699	✓ 0.11435	
thigh	1	-2.85685	2.58202	-1.11	0.2849	0.07108	
midarm	1	-2.18606	1.59550	-1.37	0.1896	0.10501	

none variables
are individually significant
at $\alpha=0.05$

★ skinfold explains
more unique variation
than thigh or arm.

“unique
contribution”

- Squared partial correlation is also called coefficient of partial determination. Has similar interpretation to coefficient of multiple determination.
- Squared partial correlation = Type II SS / (Type II SS + SSE).
- In this case, variables only explain approximately 10% of the remaining variation after the other two variables are fit.

Magnitude of effect
in SD units
* Relative Importance *

- Make regression coefficients comparable by putting all variables on the same scale.

original: β 's depend on original units X and Y
 $\Downarrow$
 * Standardizing: removes unit effect, expresses in SD units.

Standardized Regression Model

- Can reduce round-off errors in calculations
- Standardization

$$Z_Y = \tilde{Y}_i = \frac{1}{\sqrt{n-1}} \left(\frac{Y_i - \bar{Y}}{s_Y} \right) \quad \text{and} \quad Z_X = \tilde{X}_{ik} = \frac{1}{\sqrt{n-1}} \left(\frac{X_{ik} - \bar{X}_i}{s_{X_i}} \right)$$

- Puts regression coefficients in common units, such that they can be compared fairly
- A one SD change in X_i corresponds to $\tilde{\beta}_i$ SD increase in Y
- Can show

$$\underset{\text{(unitless)}}{\beta_i} = \left(\frac{s_Y}{s_{X_i}} \right) \underset{\substack{\downarrow \\ \text{unstandardized}}}{\tilde{\beta}_i}$$

Example: Body Fat

```
proc reg data=a1;  
    model fat=skinfold thigh midarm / stb;  
run;
```

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate
Intercept	1	117.08469	99.78240	1.17	0.2578	0
skinfold	1	4.33409	3.01551	1.44	0.1699	4.26370
thigh	1	-2.85685	2.58202	-1.11	0.2849	-2.92870
midarm	1	-2.18606	1.59550	-1.37	0.1896	-1.56142

skinfold has strongest
relative impact on
predicting body fat.

**Skinfold has highest standardized coefficient. Midarm does not appear to be as important a predictor. Perhaps best model includes skinfold and thigh.

Multicollinearity

- Numerical analysis problem is that the matrix $\mathbf{X}'\mathbf{X}$ is almost singular (linear dependent columns)
 - Makes it difficult to take the inverse
 - Generally handled with current algorithms
- Statistical problem: too much correlation among predictors
 - The coefficient estimation lacks interpretability.
 - Difficult to determine regression coefficients → Increased standard error
 - May not affect prediction accuracy if the testing samples follow similar multicollinear correlation.
- Want to refine model to remove redundancy in the predictors

Example

- Consider a two-predictor model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

- What is the estimate of β_1 ?
- Can show

$$b_1 = \frac{\tilde{b}_1 - \sqrt{\frac{s_Y^2}{s_{X_1}^2}} r_{12} r_{Y2}}{1 - r_{12}^2}$$

where $\tilde{b}_1$ is the estimate fitting Y vs X_1

Extreme Cases

- Consider X_1 and X_2 are uncorrelated
 - $r_{12}=0$
 - $b_1 = \tilde{b}_1$ (fitting Y vs X_1)
 - Estimator b_1 does not depend on X_2
 - Type I SS and Type II SS are the same *Each X explains its own independent portion of variation in Y.*
 - In other words, the contribution of each predictor is the same regardless of whether or not the other predictor is in the model
- Consider $X_1 = a + bX_2$ *Predictors related to one another.*
 - $r_{12} = \pm 1$ *linearly dependent*
 - Estimator b_1 does not exist
 - Type II SS are zero
 - In other words, there is no contribution of the predictor if the other predictor is already in the model

Extreme Case in SAS

- Consider the following data set

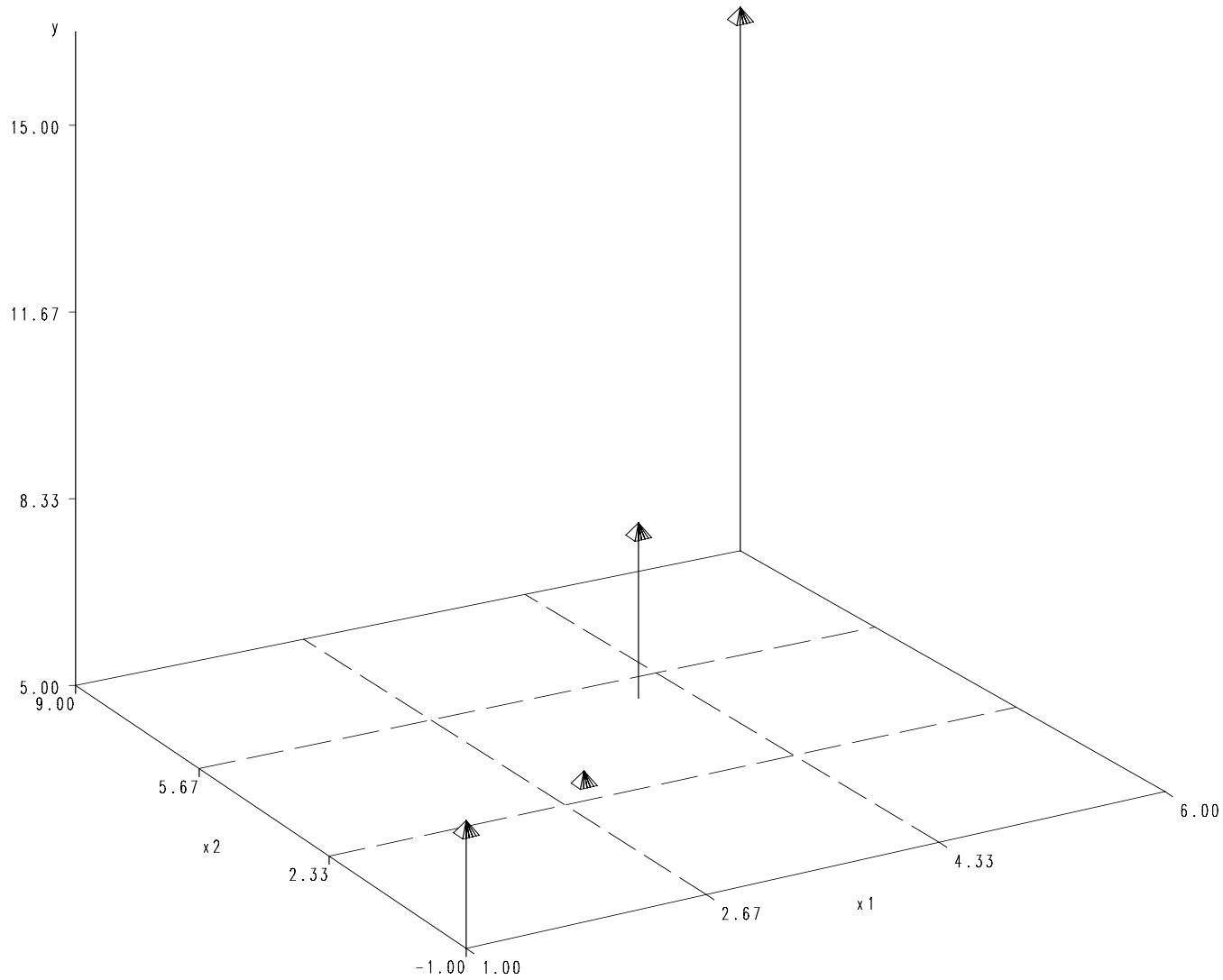
```
data a1;  
  input case x1 x2 y;  
  cards;  
  1 3 3 5  
  2 4 5 8  
  3 1 -1 7  
  4 6 9 15  
;
```

- Notice $x_2 = 2x_1 - 3$
- Will generate 3-D plot and run regression

```

/* Generate 3-D Scatterplot */
proc g3d data=a1;
    scatter x2*x1=y / rotate=30;
run;

```



```
proc reg data=a1;
    model y=x2 x1;
run; quit;
```

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	55.59211	55.59211	96.02	0.0103
Error	2	1.15789	0.57895		
Corrected Total	3	56.75000			
Root MSE		0.76089	R-Square	0.9796	
Dependent Mean		8.75000	Adj R-Sq	0.9694	
Coeff Var		8.69584			

NOTE: Model is not full rank. Least-squares solutions for the parameters are not unique. Some statistics will be misleading. A reported DF of 0 or B means that the estimate is biased.

NOTE: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

$$x1 = 1.5 * \text{Intercept} + 0.5 * x2$$

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	B	-0.65789	1.03271	-0.64	0.5893
x2	B	1.71053	0.17456	9.80	0.0103
x1	0	0	.	.	.

- In this example, no inverse exists so X_1 dropped
- In practice, we are concerned with less extremal cases
- General results still hold
 - Regression coefficients are not well estimated
 - Regression coefficients may be scientifically meaningless
 - Type I SS and II SS will differ substantially
 - R^2 and predicted values are usually ok

Prelim Diagnose: Pairwise Correlations

- Assesses “pairwise collinearity” but not complicated multicollinearity
- Consider our body fat example

```
proc reg data=a1 corr;  
    var skinfold thigh midarm fat;  
    model midarm = skinfold thigh;  
run; quit;
```

Correlation

Variable	skinfold	thigh	midarm	fat
skinfold	1.0000	0.9238	0.4578	0.8433
thigh	0.9238	1.0000	0.0847	0.8781
midarm	0.4578	0.0847	1.0000	0.1424
fat	0.8433	0.8781	0.1424	1.0000

– relatively strong correlation between thigh and skinfold.

- “MODEL midarm = skinfold thigh” reported $R^2 = 0.9904$
 - All three $\rightarrow r = \sqrt{0.9904} = .995$
 - Should not use model with all three predictors

Coefficient Estimation

- Page 284 summarizes coefficients

Variables in Model	b_1	b_2
skinfold	0.8572	-
thigh	-	0.8565
skinfold, thigh	0.2224	0.6594
skinfold, thigh, midarm	4.3340	-2.857

✓ shared information between skinfold and thigh are now divided, both explain the same portion of Y's variation.

sum is 0.86

Both alone have slope ≈ 0.85

- skinfold and thigh similar info, hence are exchangeable.
- Coeffs change when both are included (sum ≈ 0.86)
- Very dramatic change when midarm is in (but this change is still dominated by the multicollinearity)
- Reflected in std errors too

Chapter Review

- Extra Sums of Squares
- Partial correlations
- Standardized regression coefficients
- Multicollinearity
 - Effects
 - Remedies